

TECHNICAL BULLETIN



MODEL/DERIVATIVE:

100, 200, 400, 600, 800, Metro, Maestro, Montego, Mini

Bull. N°: **0003**

CDS. ref: **R3591bu**

Issue: **1**

Date: **22.02.95**

AFFECTED RANGE:

All

PROBLEM:

BRAKE JUDDER

CAUSE:

Excessive run-out on discs or rear drums where fitted, (either with original equipment parts or following fitment in service).

The primary cause of brake judder is excessive disc run out. As the maximum run out segment of disc touches the pads on each revolution, one sided wear occurs on the disc. The resulting thick/thin variation is the basic cause of judder.

ACTION:

Follow diagnostic repair chart on page 2 and refer to additional notes on pages 3 & 4

Note:

An additional A3 poster of the diagnostic repair has been included with this issue.

ADDITIONAL NOTES

ROLLER TEST

If you have access to a brake roller tester proceed as follows, if you do not have access to this equipment, please refer to ROAD TEST but note that this should not be carried out on public roads.

TEST FOR FRONT BRAKES

- 1. Position vehicle with front wheels on rollers and apply handbrake.**
- 2. Set rollers in motion and gently apply footbrake, both gauge pointers should rise in unison in a smooth steady movement in relation to pressure being applied to footbrake. Maintain a firm pressure.**
- 3. A fluctuating pointer indicates an irregularity as the disc rotates,**

this can cause brake judder. Note any suspect wheel.

NOTE: Ensure that fluctuation is observed over at least six complete revolutions of wheel.

4. Release footbrake and stop rollers.

TEST FOR REAR BRAKES

5. Position vehicle with rear wheels on rollers and chock behind front wheels. Handbrake off.

6. Set rollers in motion and gently apply footbrake, both gauge pointers should rise in unison in a smooth steady movement in relation to pressure being applied to footbrake. Maintain a firm pressure.

NOTE: Brake effort readings on rear wheels will be lower than front. Rear brakes should not lock.

7. A fluctuating pointer indicates an irregularity as the disc/drum rotates, this can cause brake judder. Note any suspect wheel.

NOTE: Ensure that fluctuation is observed over at least six complete revolutions of wheel.

8. Release footbrake and stop rollers.

ROAD TEST

This test must be carried out by an experienced driver and **must not be conducted on public roads.**

Accelerate to approx 80 kph (50 mph), declutch and lightly apply footbrake to confirm that judder is present. When judder has been reproduced, repeat the test but apply the handbrake lightly instead of the footbrake. If, during the handbrake test, no judder is evident, proceed with front wheel diagnosis path. If judder is evident during the handbrake test, proceed with the rear wheel diagnosis path.

ADDITIONAL NOTES - continued:

SKIMMING BRAKE DISCS

Engineering tests have identified that the use of a lathe to skim brake discs, (as specified in the Service Tools & Equipment programme), will ensure that disc run out is reduced to the absolute minimum.

The lower the 'final' disc run out figure, the less chance there is of brake judder/vibration returning (i.e. repeat claims).

Fitting replacement brake discs will not minimise run out as effectively as skimming the existing discs. Replacement discs should **ONLY** be fitted where the existing disc thickness is outside the Service specification.

Please note that all current Rover Cars with the exception of the Mini are fitted with discs that can be skimmed. Current Mini discs are too thin to skim , but thicker discs and modified calipers are due to be introduced during 1995.

WARRANTY CLAIMS:

Use complaint code: 6C1L - Front brake disc judder)

S.R.O information:

Time a) 70.10.39 Skim discs & check run out (wheel off & wheel on)

Time b) 70.10.12 Fit new discs & check run out (wheel off & wheel on)

Time c) 70.10.07 Fit new discs, check run out (wheel off & wheel on), skim & check run out (wheel off & wheel on)

NOTE: Times a) & c) are not applicable to Mini since disc cannot be skimmed.

Time Allowance

Mini

Time b): 2.60 hours

100/Metro

Times a): 2.60 hours

b): 1.20 hours

c): 2.80 hours

200, 400 (non vented & vented discs)

Times a): 2.60 hours

b): 1.50 hours

c): 2.80 hours

600

Times a): 2.60 hours

b): 2.10 hours

c): 3.50 hours

800

Times a): 2.60 hours

b): 1.50 hours

c): 2.40 hours

Maestro, Montego (non vented & vented discs)

Times a): 2.60 hours

b): 1.60 hours

c): 2.80 hours

Maestro Van

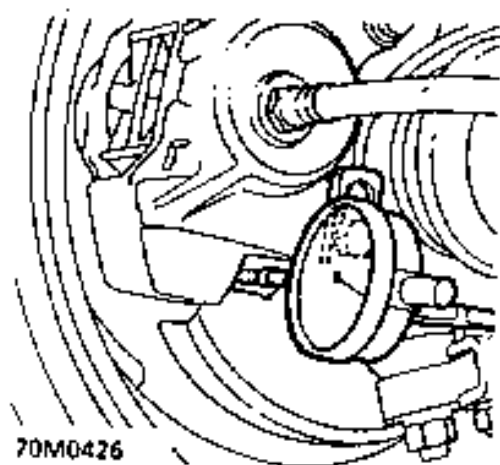
Times a): 2.60 hours

b): 3.50 hours

c): 4.20 hours

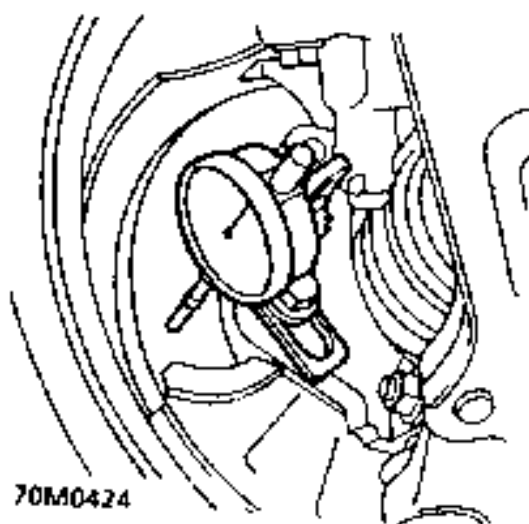
The above times relate to front disc repairs only. For rear disc/drum repairs refer to SRO manual.

NOTE: The above times supersede all previous SRO manual times, the SRO manual will be amended in due course.



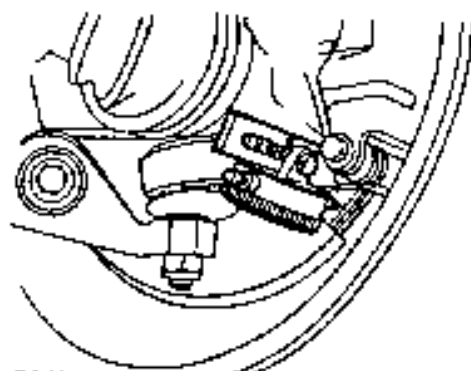
70M0426

MINI



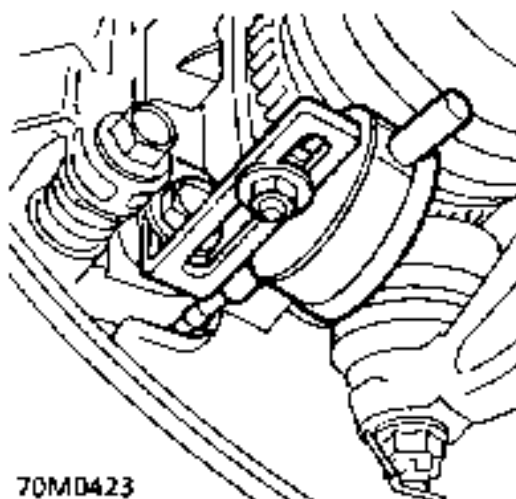
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R100/METRO



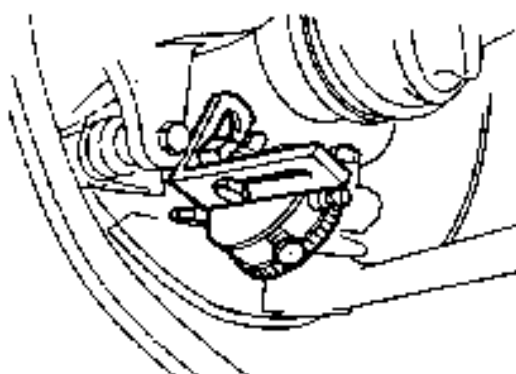
B359IRM1

R200/400



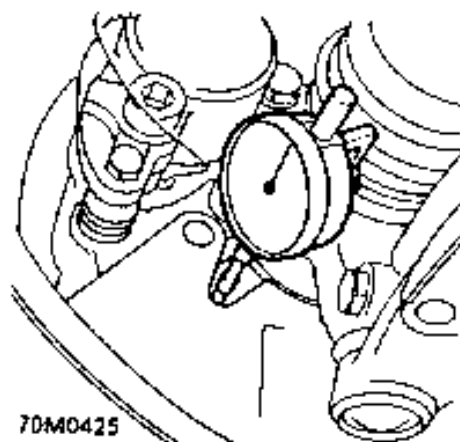
70MD423

R600



C359IRM1

R800



70M0425

MAESTRO/MONTEGO

BRAKE JUDDER REPAIR PROCEDURE

